Liquid Cooling Solutions

for High-Power Chips and Thermal Management Systems

Custom liquid cold plates | CNC machining | FSW | Vacuum brazing | Engineering support

Custom Design	**Prototype to Batch**	**Thermal Validation**

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Company Introduction

Liquid Cooling Expert for High Power Chips

Company Profile

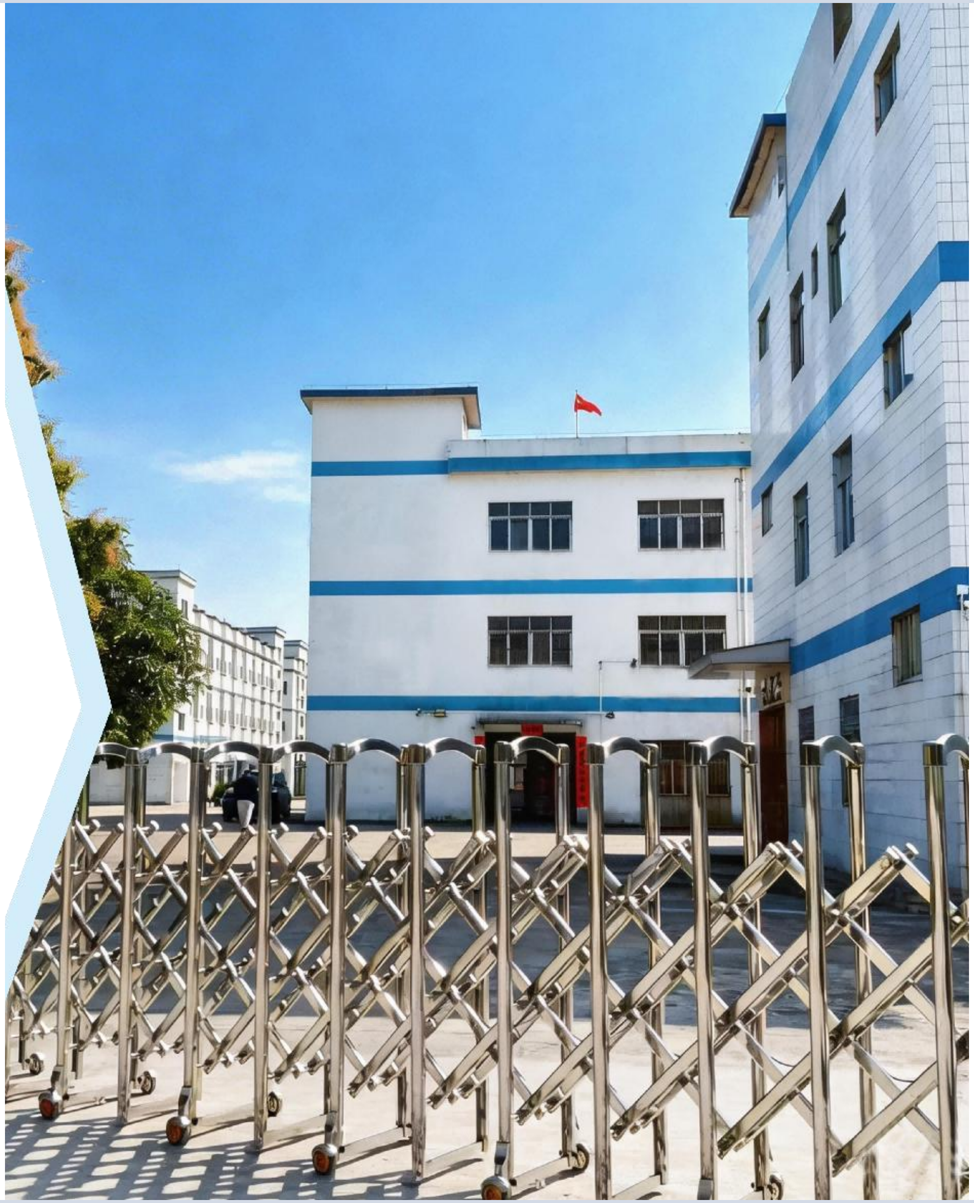
ToneCooling Technology Co., Ltd. focuses on thermal management products for high-power electronics. Our product range includes custom liquid cold plates and related cooling products for AI servers, data centers, GPUs, IGBTs, power electronics, energy storage, and new energy vehicle applications.

The company supports customer-specific projects with CNC machining, FSW, vacuum brazing, assembly, inspection, and engineering review. ToneCooling provides prototype and batch production support based on customer drawings, thermal requirements, and project validation needs.

CNC / FSW

Vacuum Brazing

Custom Engineering



Our Business

High-Performance Computing and High-Power Chip Applications

Liquid Cooling CDU and Cold Plate Development & Production

Liquid Cooling Solutions for High-Power

Patents and Certifications
Liquid Cooling Solutions for High-Power Electronics

Patent Name	Patent No.	Status	Certificate Issued	Application Date
A quick-connect and quick-disconnect device for product flow-resistance performance testing	2020216968916	Granted	✓	2020/8/14
A device for quickly filling protective gas inside a liquid cooling heat sink	2020216955992	Granted	✓	2020/8/14
Automated solid-solution device after transient liquid phase diffusion welding of aluminum alloy	2020217052837	Granted	✓	2020/8/17
Tooth-surface air-cooled heat sink with an annular pulsating heat pipe built into the substrate	2021223333279	Granted	✓	2021/9/26
Bamboo-tube water cooling plate and its coil assembly	2021223333387	Granted	✓	2021/9/26
Annular pulsating heat pipe vapor chamber	2021223706289	Granted	✓	2021/9/29
Dual-channel structure water cooling plate	2023221651844	Granted	✓	2023/8/12
High-power closed-loop air/liquid combined cooling system with six heat-source surfaces	2024202158216	Granted	✓	2024/1/30
Simulated heat source for performance testing of phase-change and liquid-cooled heat sink products	2024202158682	Granted	✓	2024/1/30
Fully automatic performance testing device with constant temperature, humidity, and flow rate	2024202929948	Granted	✓	2024/2/18
Integrated testing device for high-efficiency flow resistance, sealing, cleaning, and blow-drying	2024202930038	Granted	✓	2024/2/18
Comparative testing device for thermal performance of new materials	2024203970653	Granted	✓	2024/3/1
Efficient electrode installation device for water cooling plates	2024204647240	Granted	✓	2024/3/11
Transient liquid phase diffusion welding method for aluminum plates	2020105072054	Under Review		2020/6/5
Automated solid-solution device and method after transient liquid phase diffusion welding of aluminum alloy	2020108226995	Under Review		2020/8/17
Biomimetic blood-circulation structure water cooling plate	2023113510482	Under Review		2023/10/18
Multi-station loading and quick-insertion device for a new liquid cooling tube and barbed fitting	2023235520769	Under Review		2023/12/26
Thermal-performance simulation testing structure for high-power TEC lasers	2024213954534	Under Review		2024/6/19
Quick disassembly/assembly simulation testing structure for thermal performance of high-power double-layer lasers	2024214842790	Under Review		2024/6/27
High-power closed-loop air/liquid combined cooling system with six heat-source surfaces	2024216030453	Under Review		2024/7/9
New curved-channel and welding-surface product structure	2023235520877	Under Review		2023/12/26
Three-dimensional heat pipe heat sink	202323656193X	Under Review		2023/12/31
Quick-connect and quick-disconnect device applied to product flow-resistance performance testing	2024200888835	Under Review		2024/1/15
Thermal-performance simulation testing structure applied to high-power CPUs	2024201218258	Under Review		2024/1/18

Certified to ISO 9001.

Design and R&D and welding

Liquid Cooling Expert for High Power Chips

Customized Service
Liquid Cooling Solutions for High-Power Electronics

- **Customers provide the design conditions, and our engineering team develops a thermal solution based on the application requirements. After testing confirms that the solution meets customer requirements, the project can move toward small-batch validation and mass production.**

Boundary Conditions
- 3D structure
- Heat-source parameters
- Fluid parameters
- Performance

Design Verification
- Structural design
- Simulation analysis
- Sample preparation
- Test verification
- Small-batch validation

Mass Production
- Supplier qualification audit
- Mass production order

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After-Sales Service

Liquid Cooling Solutions for High-Power Electronics

- Dedicated support is provided for key customers, including efficient online communication and on-site service response when required. Warranty or replacement support can be provided for confirmed quality issues according to the agreed terms.

- **Effective**

  Fast response for quality issues and installation instructions

  Professional

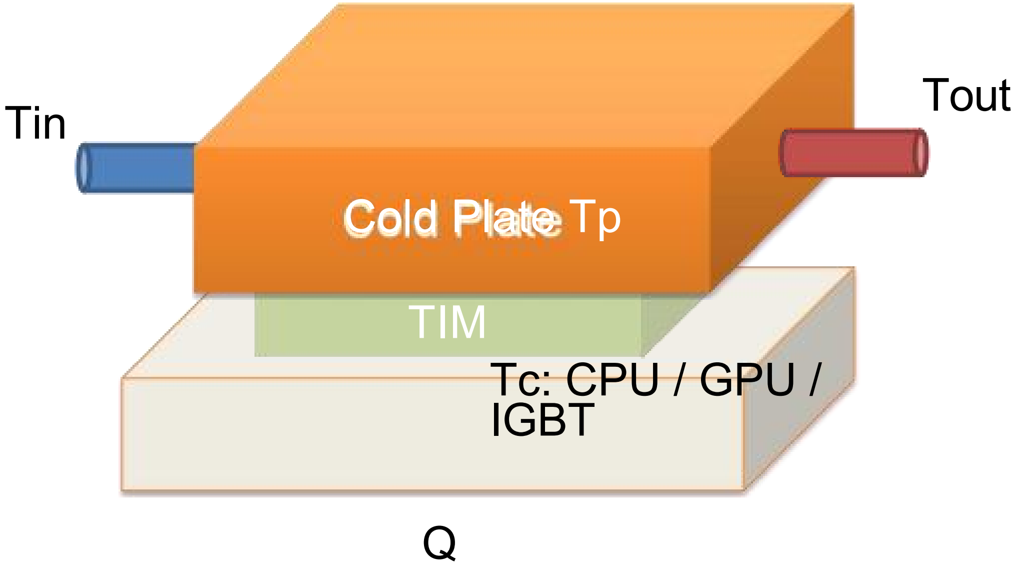
  Experienced engineering team with years of project experience

  Diligence

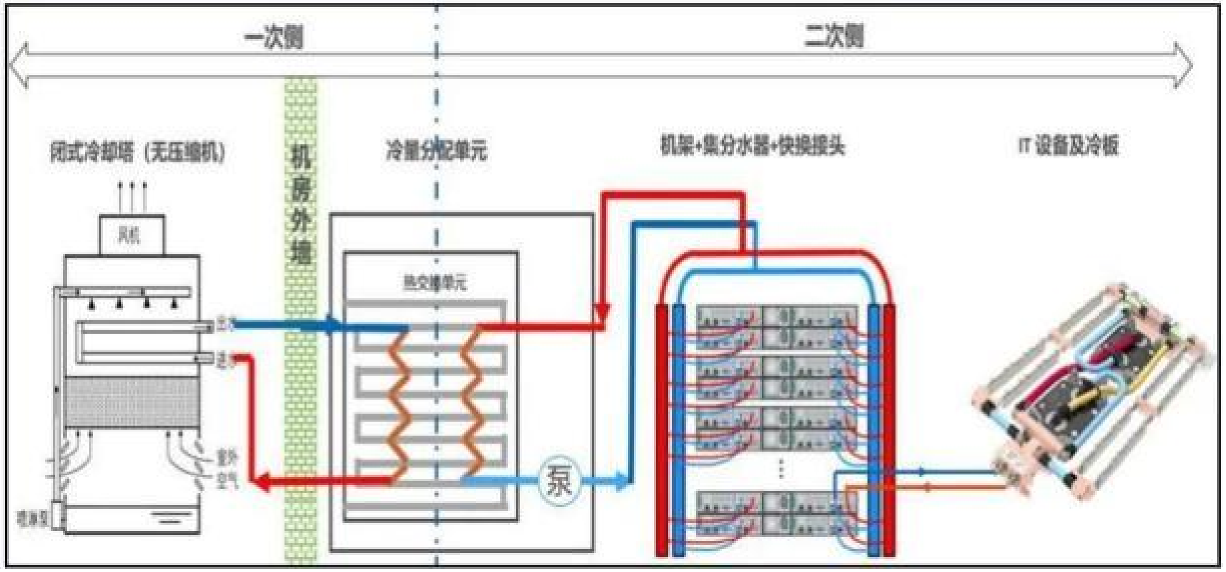
  Comprehensive service provided for our clients

Design Input

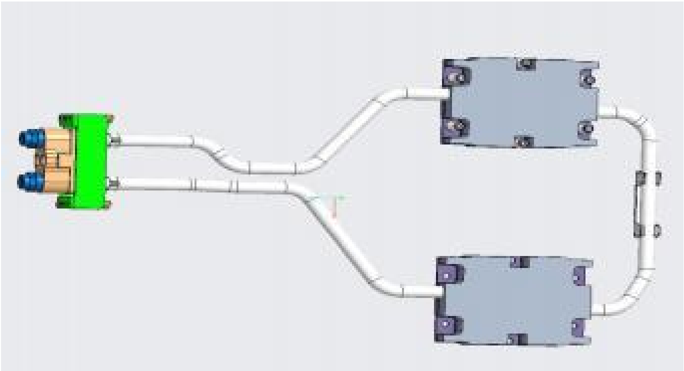
- 3D structure and connector limitations
- Coolant, flow rate, and hydraulic resistance
- TDP and thermal resistance
- Inlet and outlet routing



Cooling Plate and Heat Source



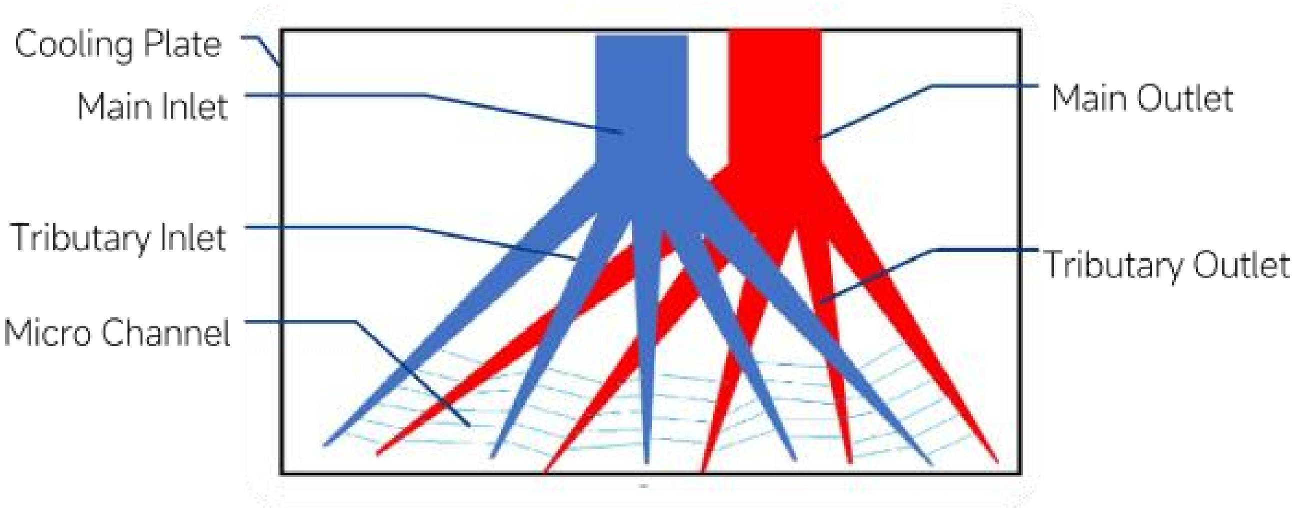
Link-level Graph of Liquid Cooling Plate


Cooling Plate and Sets

Liquid Cooling Expert for High Power

Biomimetic Channel Structure
- Hierarchical flow path: main inlet/outlet, branch channels, and microchannels
- Designed to support high heat-flux dissipation with lower flow resistance
- Example result: 1,000 W cooling target, 2 L/min flow rate, approximately 7,500 Pa flow resistance, and 14.5 deg C temperature rise under tested conditions

	Biomimetic Cold Plate	Traditional Cold Plate
Temperature field Flow	Good temperature uniformity	Big differences between in & out
Resistance	Low	High
Heat flux	>200W/CM2	<100W/CM2

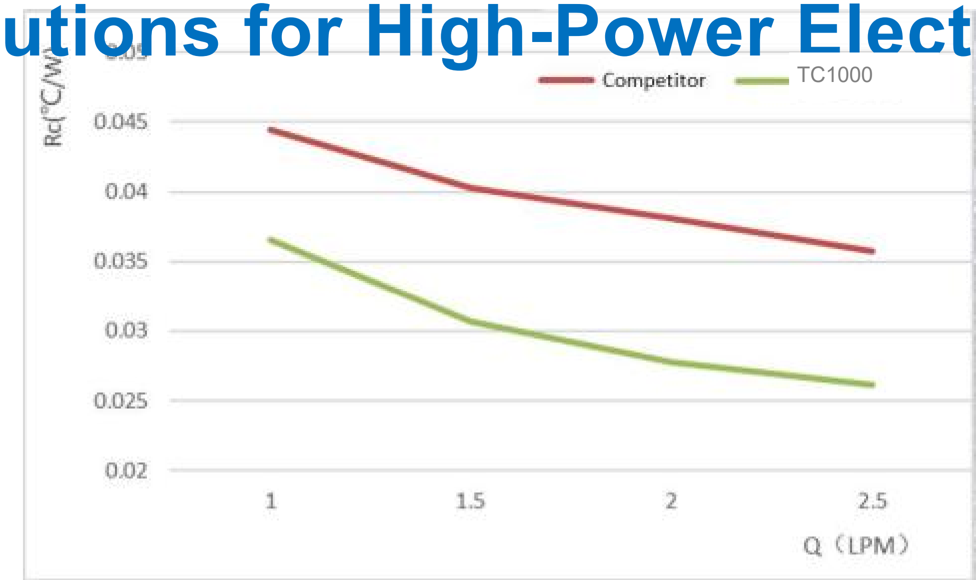


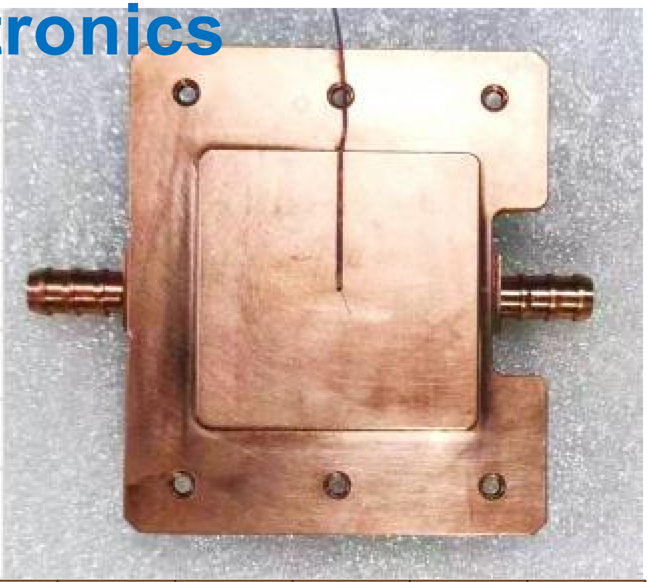
Liquid Cooling Solutions for High-Power Electronics

1000 W Liquid Cooling Plate
Liquid Cooling Solutions for High-Power Electronics

Reference Comparison
- 10 deg C lower than the reference solution
- THS1000 for GPU / ASIC, 1,000 W
- Copper cold plate with LM-188J TIM
- Pressure drop: 8 kPa at 2 LPM
- Dimensions: 83 x 70 x 16.3 mm

Requirements:
- Flow rate: 2 LPM
- Temperature rise: < 15 deg C
- Pressure drop: < 10 kPa





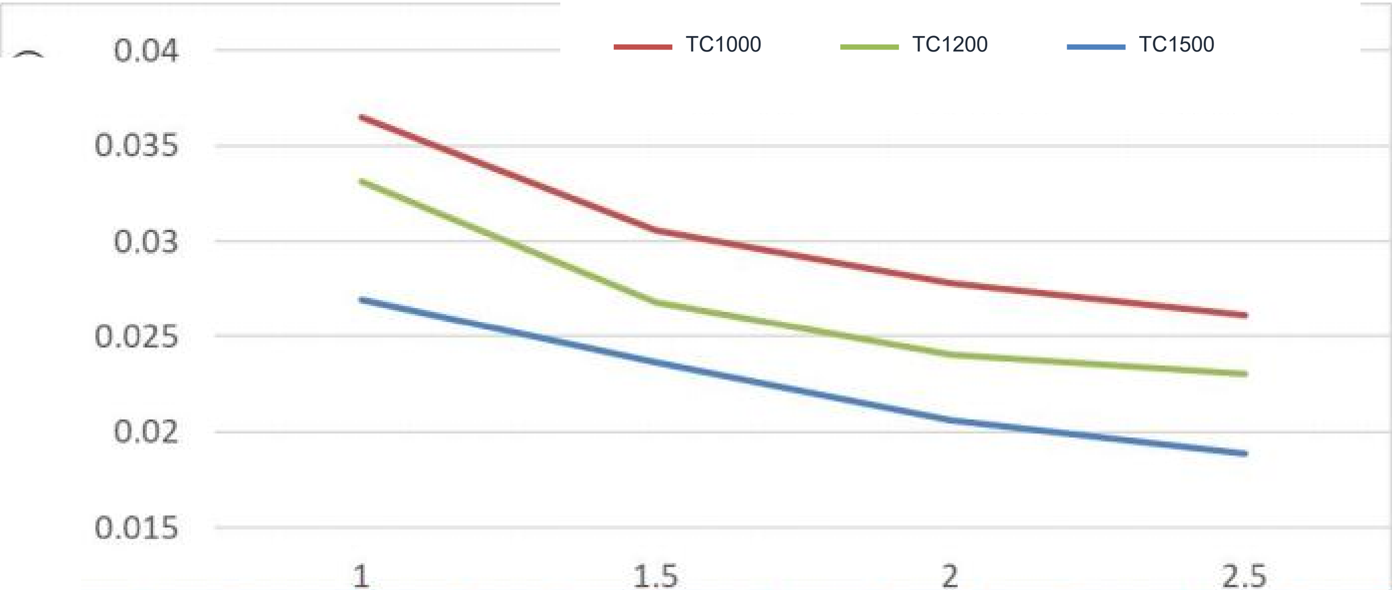
Sample	Power (W)	Q (LPM)	Tw-in (deg C)	Tw-out (deg C)	DeltaTw (deg C)	Tp (deg C)	Tc (deg C)	DeltaTp (deg C)	DeltaTc (deg C)	Rp(deg C/W)	Rc(deg C/W)	dp (kPa)
Competitive products	1003.7	1	29.3	41.9	12.6	64.14	73.87	34.84	44.57	0.0347	0.0444	1.35
	1000.3	1.5	29.7	38.2	8.5	60.3	70.02	30.6	40.32	0.0306	0.0403	2.78
	1000.9	2	30.3	37.3	7	58.29	68.45	27.99	38.15	0.0280	0.0381	4.78
	1004	2.5	29.1	35	5.9	55.68	65	26.58	35.9	0.0265	0.0358	7.27
TC1000	1005.4	1	27.8	41.5	13.7	49.68	64.52	21.88	36.72	0.0218	0.0365	2.01
	1003.6	1.5	27.5	36.5	9	45.39	58.24	17.89	30.74	0.0178	0.0306	4.37
	1000.1	2	28.6	35.5	6.9	43.12	56.41	14.52	27.81	0.0145	0.0278	7.51

TC1200 / TC1500

TC1200 and TC1500 are designed for higher cooling capacity while maintaining comparable flow resistance.

TC1500 supports a 1500 W cooling target under the tested condition. The measured thermal resistance can reach Rc = 0.0192 K/W and Rp = 0.0097 K/W.

Liquid Cooling Solutions for High-Power Electronics

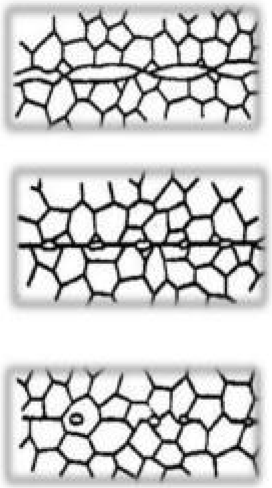


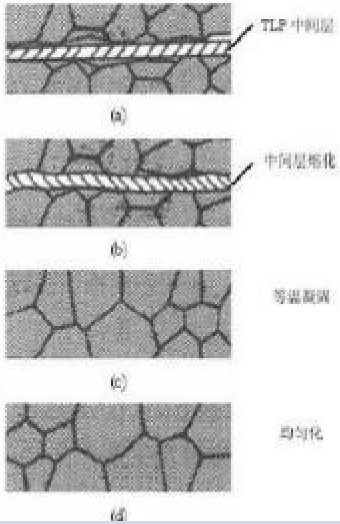
ple	Power (W)	Q (LPM)	Tw-in (℃)	Tw-out (℃)	△Tw (℃)	Tp (℃)	Tc (℃)	△Tp (℃)	△Tc (℃)	Rp(℃/W)	Rc(℃/W)	dp (kpa)
TC1200 (LM-188J)	1202. 0	1	29. 2	45. 2	16	56. 24	69. 04	27. 04	39. 84	0.0225	0. 0331	1. 6
	1203. 3	1. 5	29. 6	40. 4	10. 8	49. 54	61. 87	19. 94	32. 27	0.0166	0. 0268	3. 79
	1201. 7	2	30. 1	38. 2	8. 1	47. 37	58. 97	17. 27	28. 87	0.0144	0. 024	6. 41
	1206. 4	2. 5	30. 2	36. 8	6. 6	46. 64	57. 97	16. 44	27. 77	0.0136	0. 023	10. 43
TC1500 (LM-188J)	499.30	1	29.6	50.0	20.4	54.48	70.39	24.88	40.79	0.0166	0.0272	2. 4
	499.30	1. 5	30.0	43.4	13.4	50.29	65.71	20.29	35.71	0.0135	0.0238	4. 7
	499.30	2	29.3	39.6	10.3	46.90	61.61	17.60	32.31	0.0117	0.0215	6. 86
	499.30	2. 5	30.5	38.8	8.3	45.10	59.23	14.60	28.73	0.0097	0.0192	10. 82

Patented Welding Technology

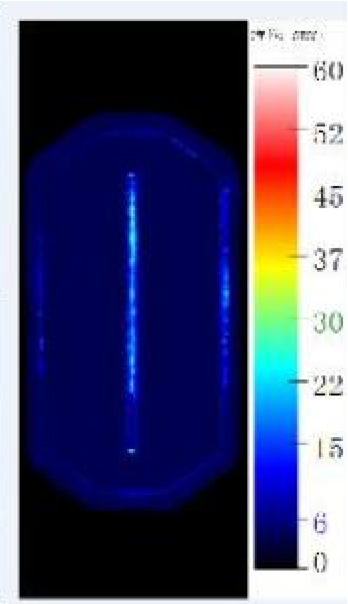
- The welding process is critical to thermal resistance and reliability. ToneCooling developed a transient liquid phase (TLP) welding process supported by specialized welding equipment.
- Compared with conventional vacuum brazing, TLP welding can support higher process efficiency, stronger bonding strength, and improved yield under the applicable design conditions.

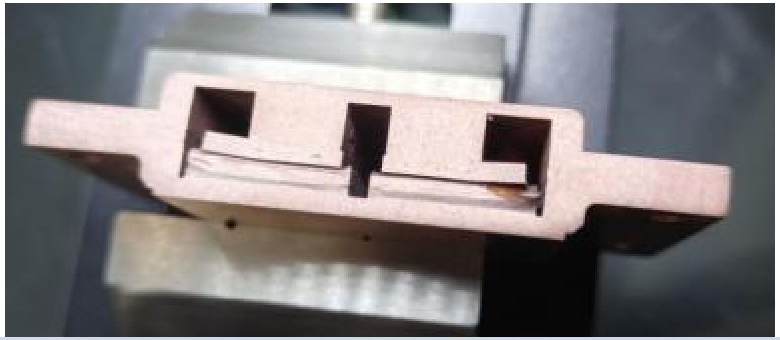
Liquid Cooling Expert for High Power Chips





	TLP	Vacuum Brazing
Efficiency	5 min/pc	1 h/pc
Strength	0.85	0.75
Yield	>99%	>90%

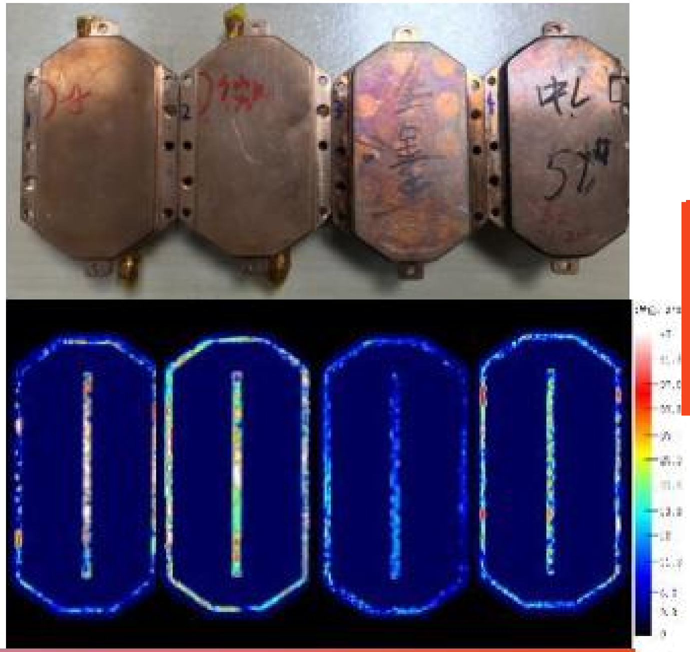




Reliable Welding Process
Liquid Cooling Solutions for High-Power Electronics

- Ultrasonic C-scan results showed improved welding quality and process reliability.

Passed - ToneCooling Sample

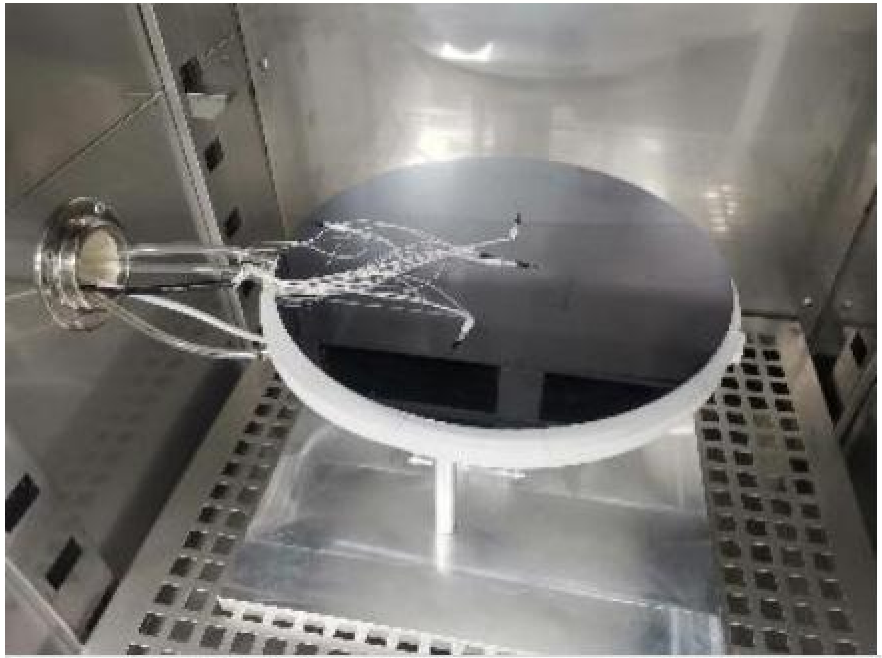


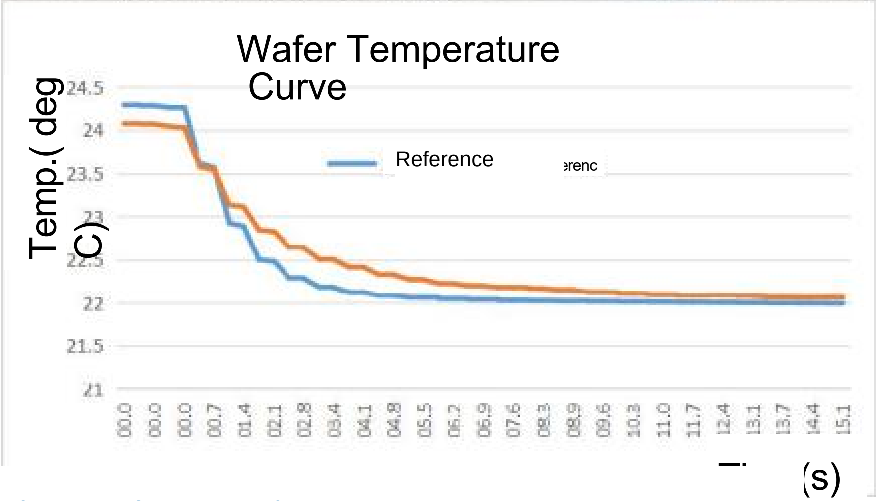
Not Passed

Reference Sample

*Defects are marked in

Project Summary
Customer: Customer project
Supplier: ToneCooling
Project No.: YDZX20223100004016
Scope: Develop one liquid cooling unit for wafer temperature control.
Key targets: diameter 310-320 mm; flatness <= 10 um; profile height <= 50 um; target temperature 22 +/- 0.07 deg C; time to target <= 7.3 s.
Deliverables: cooling units, design documents, drawings, simulation report, and test report.



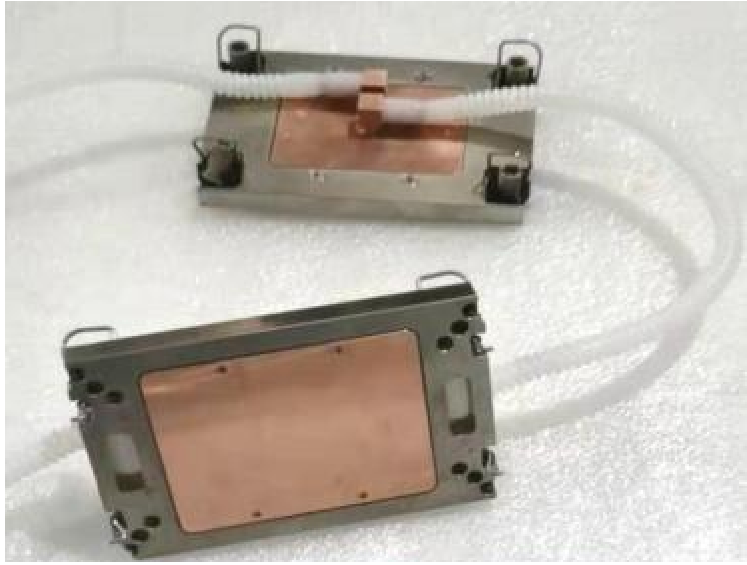


Liquid Cooling Solutions for High-Power Electronics

Intel CPU Liquid Cooling Plate

Name: Intel CPU Cooling Plate
Source: Eagle Stream CPU
Power: 350 W x 2
Material: Copper
Welding: TLP
Coolant: 25% PG
Flow Resistance: 16 kPa at 1 LPM
Dimensions: 118 x 78 x 10 mm
Requirements:
1. Flow rate: 1-1.25 LPM
2. Inlet temperature: 40 deg C
3. CPU die: < 53 deg C
4. HBM die: < 51 deg C
5. CPU temperature difference: < 2 deg C

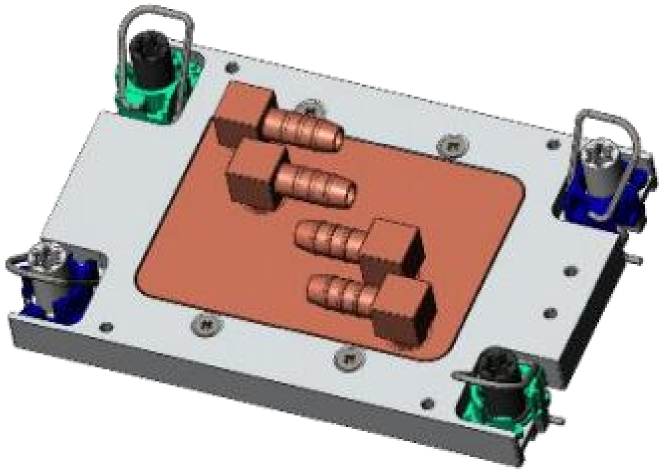
Liquid Cooling Solutions for High-Power Electronics



Intel CPU

Name: Dual-Channel Cold Plate for Server
Source: Eagle Stream CPU
Power: 350 W x 2
Material: Copper
Welding: TLP
Coolant: 25% PG
Flow Resistance: 6 kPa at 1 LPM
Dimensions: 118 x 78 x 16.5 mm
Requirements:
1. Flow rate: 1 LPM
2. Inlet temperature: 40 deg C
3. Two independent flow paths; either path can meet the heat dissipation requirement during operation.
4. CPU die: < 53 deg C under single-channel operation

Liquid Cooling Solutions for High-Power Electronics

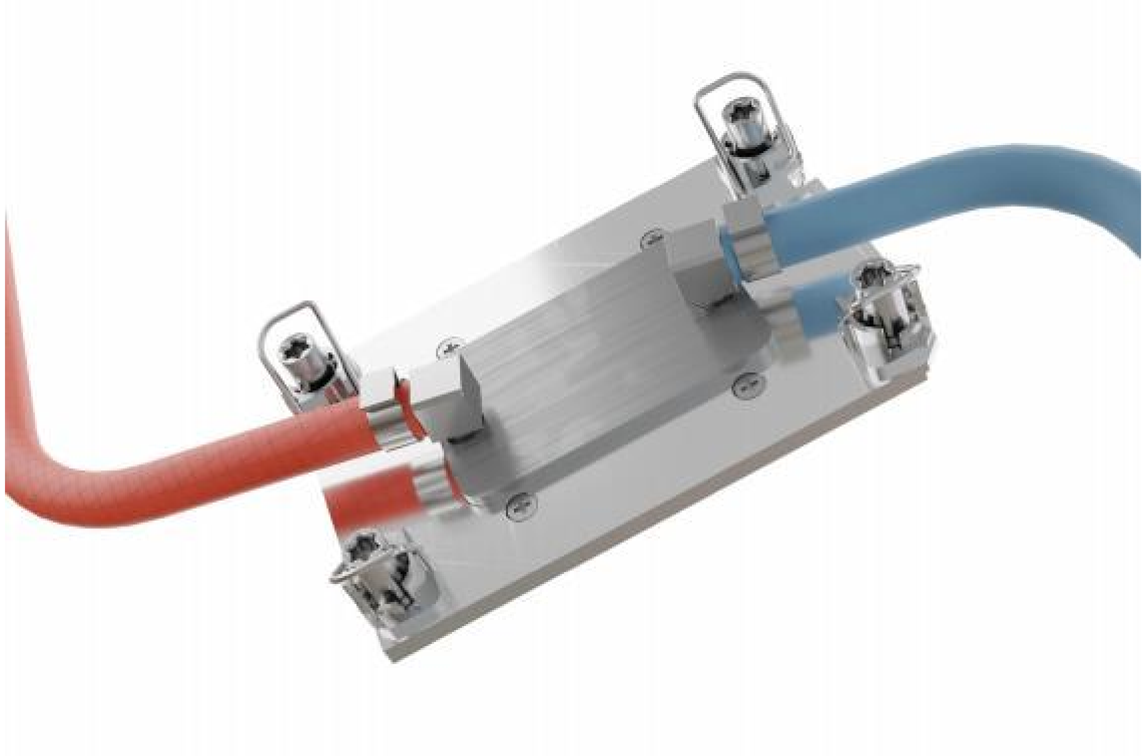


Intel Eagle Stream OC cold plate

- CPU TDP: 800 W
- Coolant input: 25% EGW, 40 deg C, 3 L/min
  Hydraulic resistance: 25 kPa
- Thermal grease: LM888W
- Measured temperature: 10 deg C lower than the reference solution
- Liquid Cooling Solutions for High-Power Electronics

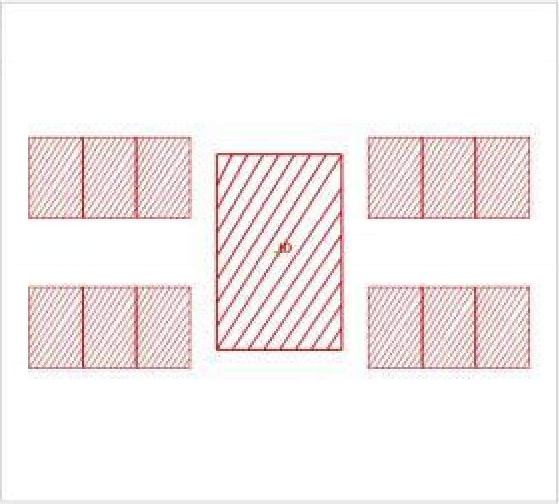
Intel Birch Stream Liquid Cooling Plate

Heat Source: Birch Stream CPU
Heat Dissipation Capacity: 350 W
Material: Copper T2
Process: TLP welding
Working Fluid Conditions: pure water, 48 +/- 2 deg C, 1.5 L/min
Pressure Drop: < 15 kPa
Surface Temperature: < 58 deg C
Pressure-Bearing Capacity: 1.0 MPa
Design Requirements:
Water supply temperature: 48 deg C
Bonding surface temperature: < 60 deg C
Flow resistance: < 15 kPa
Liquid Cooling Solutions for High-Power Electronics

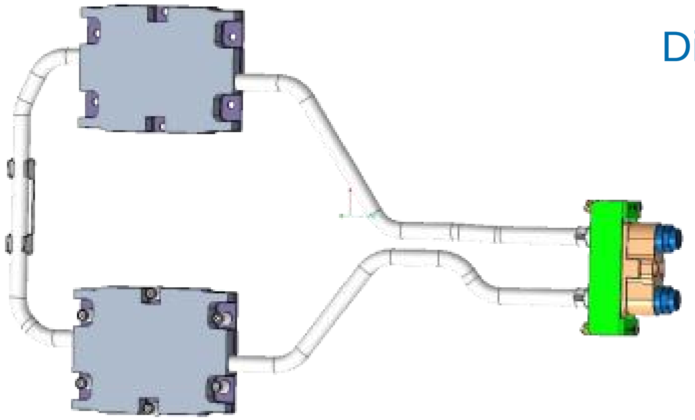


AMD CPU Liquid Cooling Plate

Name: AMD CPU Cold Plate
Source: AMD SP5 CPU
Power: 500 W x 2
Material: Copper
Welding: TLP
Coolant: 25% PG
Flow Resistance: 44 kPa at 2 LPM
Dimensions: 116 x 92 x 18 mm
Requirements:
1. Flow rate: 2 LPM
2. Inlet temperature: 40 deg C
3. Plate temperature: < 52 deg C
4. Flow resistance: < 50 kPa
Liquid Cooling Solutions for High-Power Electronics



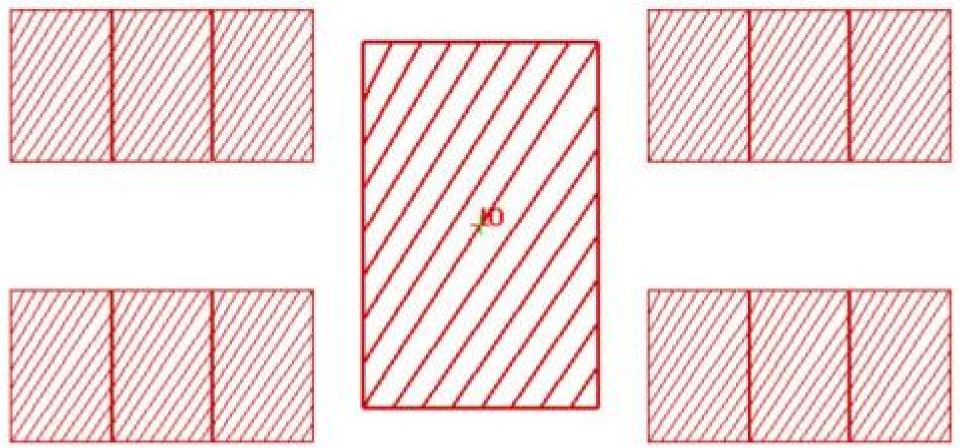
Heat Source / **e /**
Temperature
Distribution



AMD SP5 Cold Plate

- CPU TDP: 500 W
- Coolant input: pure water, 42 deg C, 1 L/min
  Coolant output: 49.1 deg C
- Four-die temperature uniformity: 52 deg C

Chip Heat-Source Distribution



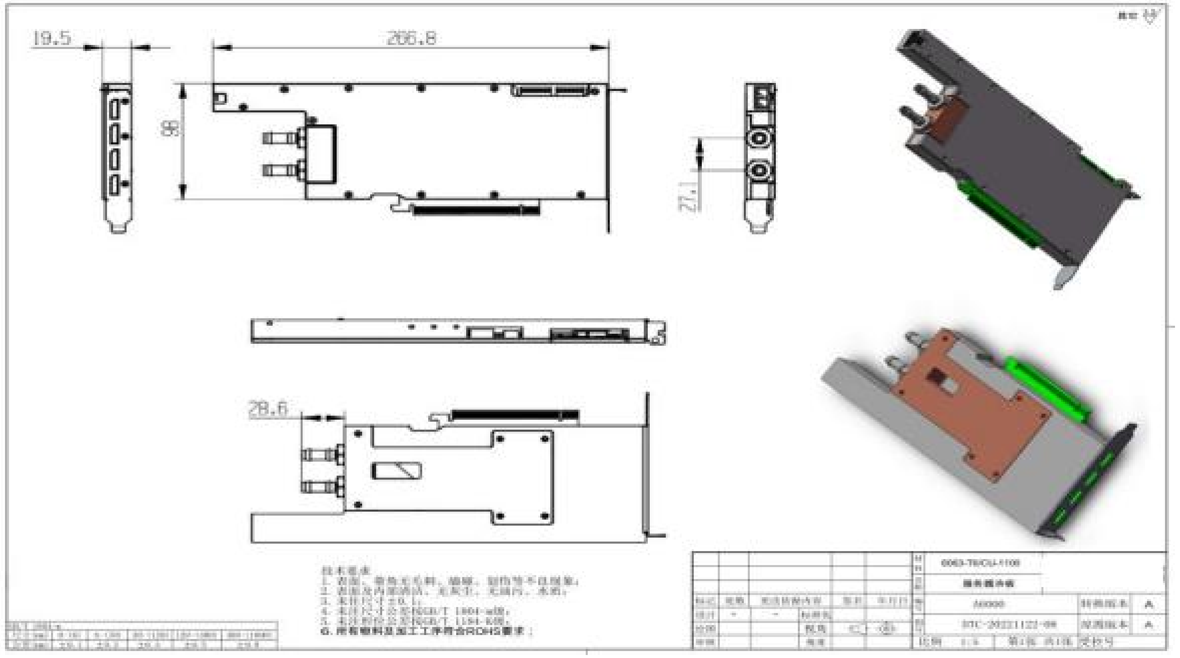
Liquid Cooling Expert for High Power Chips

Nvidia GPU Cold Plate

Name: A6000 Cold Plate
Source: NVIDIA A6000 GPU
Power: 300 W
Material: Copper / Aluminum
Welding: TLP
Coolant: 34% glycol
Pressure Drop: 2 kPa at 1 LPM
Dimensions: 266.8 x 98 x 19.5 mm
Requirements:

1. Inlet temperature: <= 50 deg C
2. Plate temperature: <= 62 deg C
3. Pressure drop: <= 2 kPa
4. Dimensions: 270 x 100 x 20 mm3

Liquid Cooling Solutions for High-Power Electronics



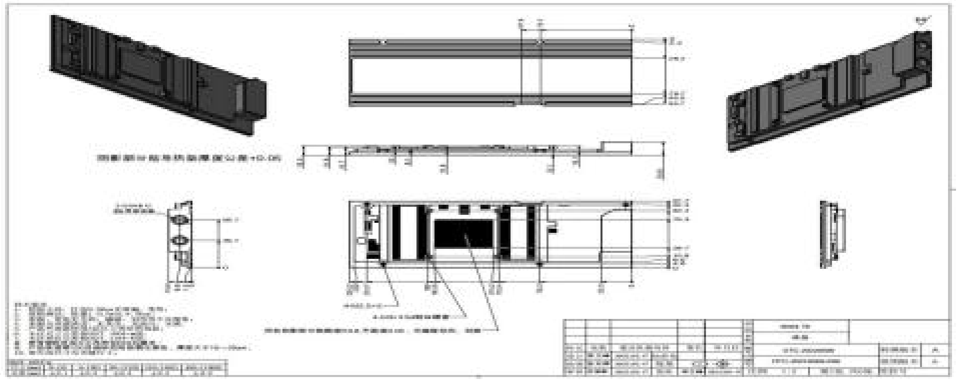
Liquid Cooling Solutions for High-Power Electronics

NVIDIA GPU Cold Plate

Product Name: A100 Cold Plate
Heat Source: NVIDIA A100 GPU
Power Dissipation: 350 W
Material: Aluminum
Welding Process: Friction Stir Welding (FSW)
Coolant: 57% glycol-water mixture
Pressure Drop: 8.5 kPa at 1.5 L/min
Dimensions: 420 x 100 x 20 mm

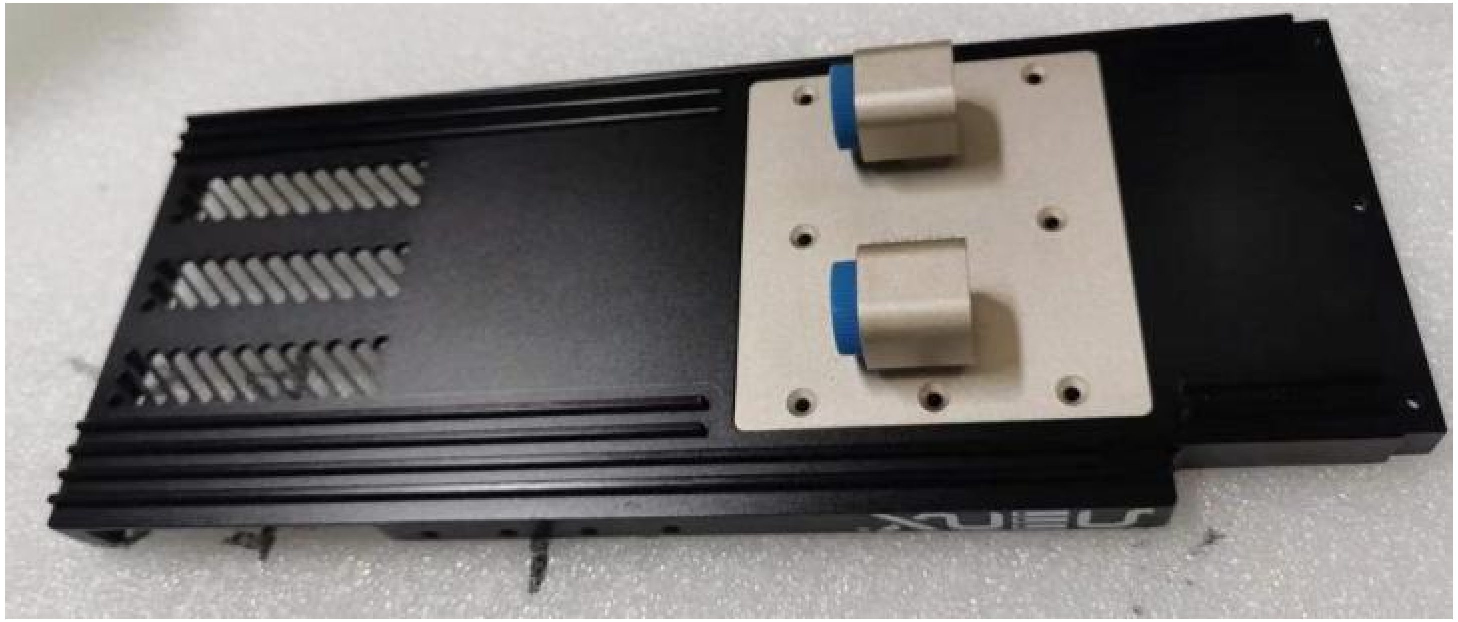
Design Requirements:

Coolant inlet temperature: <= 42 deg C
Cold plate temperature: <= 55 deg C
Pressure drop: <= 25 kPa
Dimensions: 420 x 100 x 20 mm



Nvidia GPU Cold Plate

Product Name: 3080 Cold Plate
Heat Source: NVIDIA RTX 3080 GPU
Power Dissipation: 368 W
Material: Copper / Aluminum
Welding Process: Transient Liquid Phase Bonding (TLP)
Coolant: 30% glycol-water mixture
Pressure Drop: 4 kPa at 1 L/min



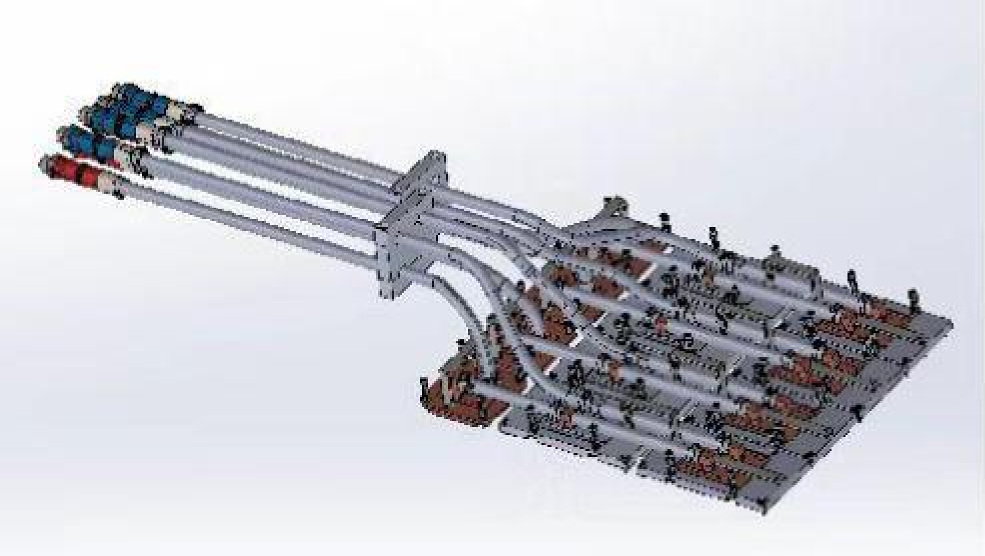
Design Requirements:

Coolant inlet temperature: ≤ 50°C
Cold plate temperature: ≤ 61°C
Pressure drop: ≤ 5 kPa

Liquid Cooling Expert for High Power Chips

Liquid Cooling Plate for NVIDIA AI GPU (8 PCS)

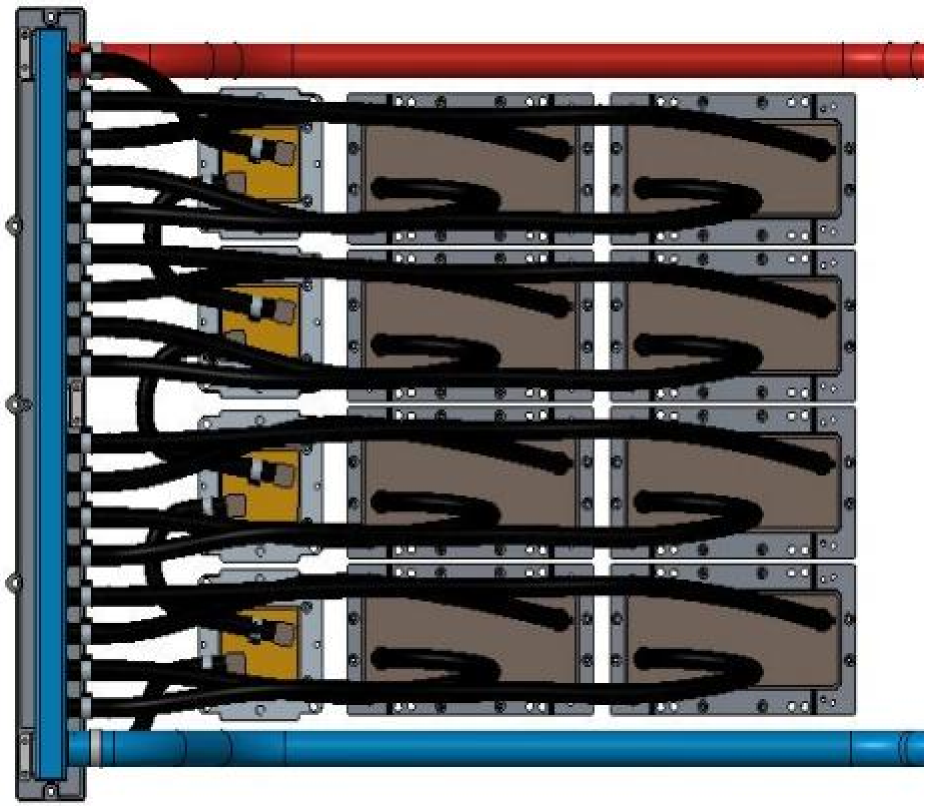
Heat Source: NVIDIA H100
Heat Dissipation: 700 W x 8 + 186 W x 4
Material: Oxygen-free copper T2
Process: TLP welding
Flow Rate Requirement: 10 L/min
Liquid Medium: Deionized water
Overall Dimensions: 906 x 331 x 139.6 mm
Flow Resistance: < 40 kPa
Mating Surface Temperature: Tcase < 65 deg C
Application: Data center
Surface Requirements: mating surface without treatment, Ra 0.8; visible surface painted black; aluminum parts black hard anodized.
Liquid Cooling Solutions for High-Power Electronics



*Customized design based on customer requirements.

SXM5 Die Liquid Cooling

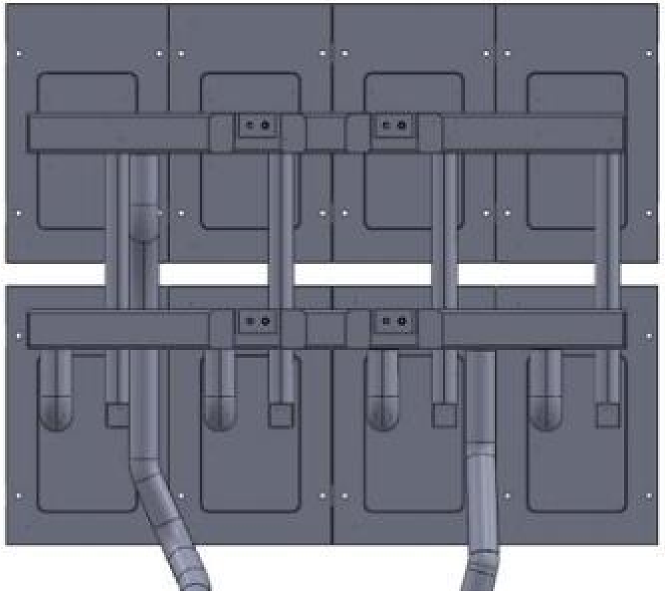
NO.	Project	Parameter
1	Platform	NV H100
2	TDP	GPU=700*8, NV Switch=134W*2， 156W*2； NV VRswitch=15.5W*8
3	Coolant	PG25
4	Input	40 deg C
5	Velocity	8~10L/min
6	Flow Resistance	30kPa



Liquid Cooling Expert for High Power Chips

1000 W x 8 Chip Cooling Plate

- CPU TDP: 1000 W x 8
- Coolant: PG25
- Input temperature: 40 deg C
- Flow rate: 12 LPM
- Configuration: four TC1200 units and four TC1500 units per set; two sets connected in series
- Temperature difference: +/-3 deg C
- Thermal grease: LM888W
- Surface temperature: 64.3 deg C / 66.7 deg C



Liquid Cooling Expert for High Power Chips

Nvidia GPU Cold Plate

Name: 4090 Cold Plate
Source: NVIDIA 4090 GPU
Power: 450 W
Material: Copper / Aluminum
Welding: TLP
Coolant: 34% glycol
Pressure Drop: 2.5 kPa at 1 LPM

Requirements:
1. Inlet temperature: <= 50 deg C
2. Plate temperature: <= 62 deg C
3. Pressure drop: <= 2.5 kPa
Liquid Cooling Solutions for High-Power Electronics

Liquid Cooling Solutions for High-Power Electronics

NVIDIA GPU Cold Plate

Product Name: 4090 Cold Plate
Heat Source: NVIDIA RTX 4090 GPU
Power Dissipation: 450 W
Material: Copper / Aluminum
Welding Process: Transient Liquid Phase Bonding (TLP)
Coolant: 20% propylene glycol-water mixture
Pressure Drop: 8 kPa at 0.5 L/min

Design Requirements:

Coolant inlet temperature: <= 38 deg C
Cold plate temperature: <= 63 deg C
Pressure drop: <= 8 kPa

Nvidia 4090 Liquid Cooling Server

Single GPU with Air Cooling

4090 Liquid Cooling Server (8

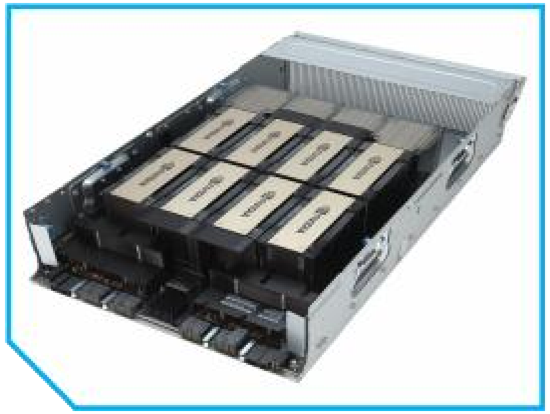
Liquid Cooling Solutions for High-Power Electronics
4090 Computing Server with 8 GPUs

NO.	Entry	Quantity
1	GPU Plate	8pcs/set*2500pcs=20,000pcs
2	Connector	16pcs/set*2500pcs=40,000pcs
3	Manifold	2pcs/set*2500pcs=5,000pcs
4	Manifold Joint	20pcs/set*2500pcs=50,000pcs
5	Pipe	16pcs/set*2500pcs=40,000pcs

Comparison of NVIDIA 4090 Systems: Liquid Cooling vs. Air Cooling

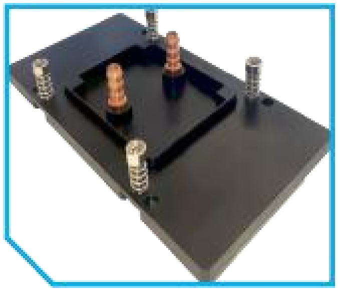
	Air Cooling 4090 Server	Liquid Cooling 4090 Server	
Single Slot Occupancy	4 Slot (<=4 GPU in single server)	1-2 Slot (8-16 GPU in single server)	More Space / Better Scalability
Fault Rate	7-8% annually	0-1% annually	Lower Failure Rate
PUE	>1.4	<1.1	Higher Energy Efficiency / Lower Electricity Cost
Clock Frequency	2280 MHz	2520 MHz	Higher Clock Frequency
Chip Temperature	Relatively High	Relatively Low	Better Cooling Capacity
Chip Lifetime	Device lifetime decreases as temperature rises.	According to the Arrhenius equation, device lifetime may be reduced as operating temperature rises.	Longer Lifetime

Nvidia AI GPU Liquid Cooling Server

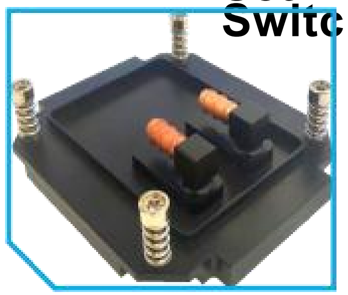


AI GPU Server

Complete Cold Plate Set



GPU Plate



Switch Plate

Single Cold Plate

Liquid Cooling Solutions for High-

Computing Server with 8 pcs of AI GPU

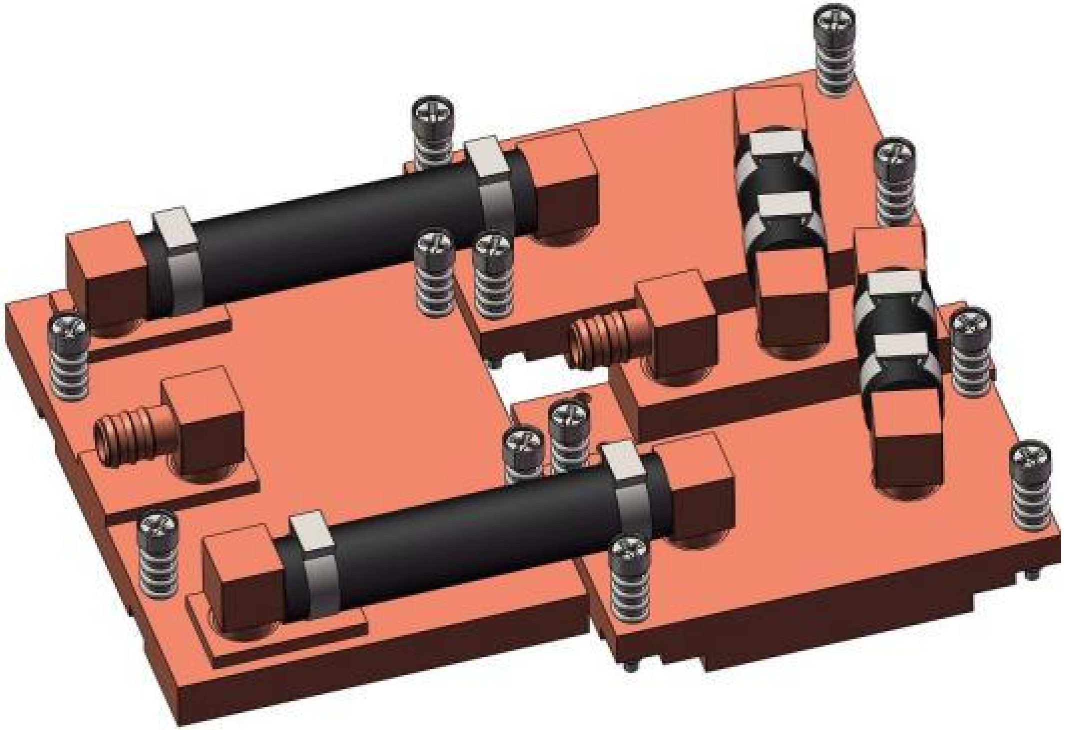
NO.	Entry	Quantity
1	CPU Plate	2pcs/set*800sets=1600pcs
2	CPU Pipe	4pcs/set*800sets=3200pcs
3	GPU Plate	8pcs/set*800sets=6400pcs
4	GPU Pipe	16pcs/set*800sets=12800pcs
5	NVLink Plate	4pcs/set*800sets=3200pcs
6	NVLink Pipe	8pcs/set*800sets=6400pcs
7	Main Pipe	8pcs/set*800sets=6400pcs
8	Detection Line Leakage	4pcs/set*800sets=3200pcs

Comparison of NVIDIA AI GPU Systems: Liquid Cooling vs. Air Cooling

	Air-Cooled AI GPU Server	Liquid Cooling AI GPU Server	
Chassis Cleanliness	Lower risk of dust-related static electricity damage	Higher cleanliness after air-cooling units are removed	Higher Cleanliness
GPU Failure Rate	7-8% annually annually	0-1%	Lower Failure Rate
PUE	>1.4	<1.1	Higher Energy Efficiency / Lower Electricity Cost
Computational Efficiency	Low	High	Higher CE
Chip Temperature.	Relatively High	Relatively Low	Better Cooling Capacity
Chip Lifetime	Device lifetime decreases as temperature rises.	According to the Arrhenius equation, device lifetime may be reduced as operating temperature rises.	Longer Lifetime

Nvidia GB200 Liquid Cooling Plate

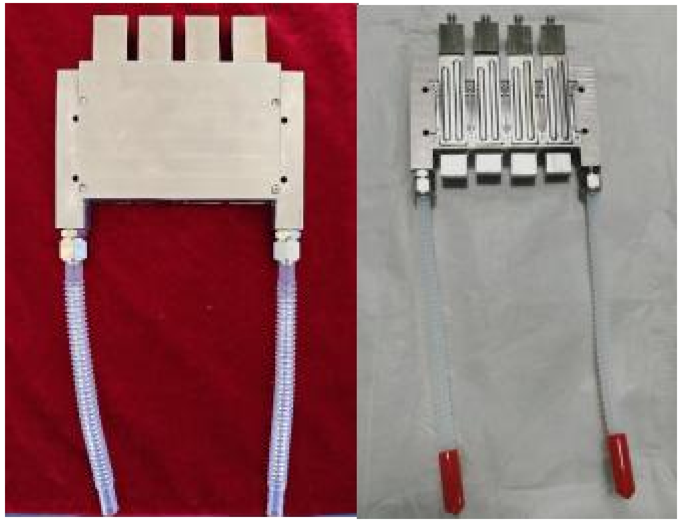
Heat Source: GB200 GPU
Heat Dissipation: 1200 W + 1200 W + 200 W
Material: Oxygen-free copper T2
Process: TLP welding
Working Medium: pure water, 25 +/- 2 deg C, 4.4 L/min
Pressure Drop: 29 +/- 2 kPa
Surface Temperature: < 55 deg C
Pressure Bearing: 0.6 MPa
Design Requirements:
Water supply temperature: 25 deg C
Bonding surface temperature: < 70 deg C
Flow resistance: < 40 kPa



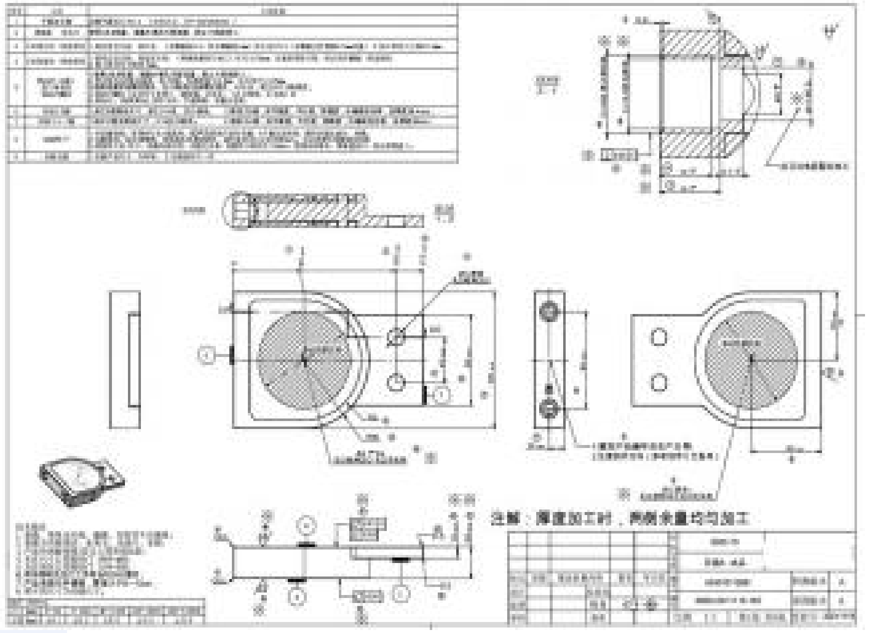
Liquid Cooling Expert for High Power Chips

Optical Module Liquid Cooling Plate

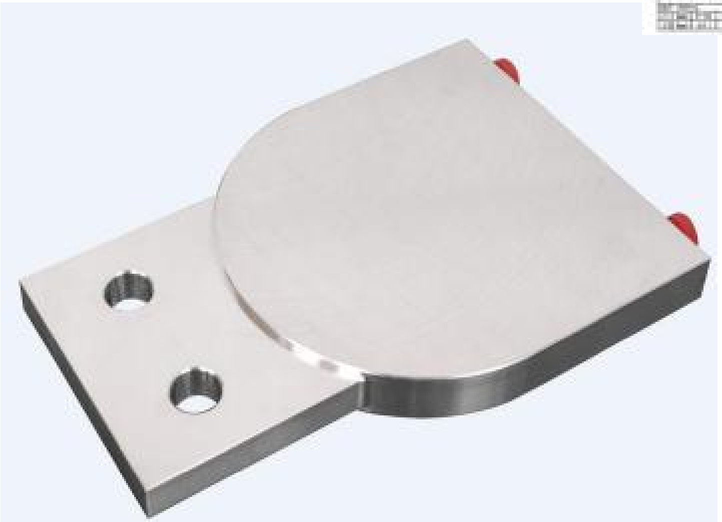
Product Name: Optical Module Liquid Cooling Plate
Heat Source: 4015 mm Optical Module
Heat Dissipation: 25 W x 16 = 400 W
Materials: Cold plate: copper; liquid distributor: stainless steel
Manufacturing Process: Ultrasonic Welding
Working Fluid: PG25
Overall Dimensions: 106 x 80 x 12.9 mm for four optical modules; length varies according to the number of modules
Pressure Drop Requirement: < 25 kPa; designed value: 7.9 kPa
Flow Rate: 1.2 L/min
Coolant Supply Temperature: 40°C
Operating Temperature Range: 5–70°C
Surface Treatment: Electroless nickel plating, according to final requirements



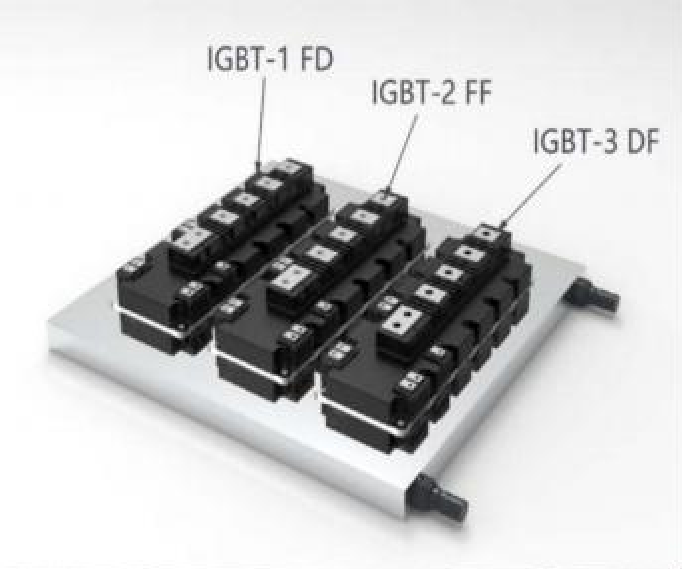
Cold Plate for Ultra-high Voltage Converter



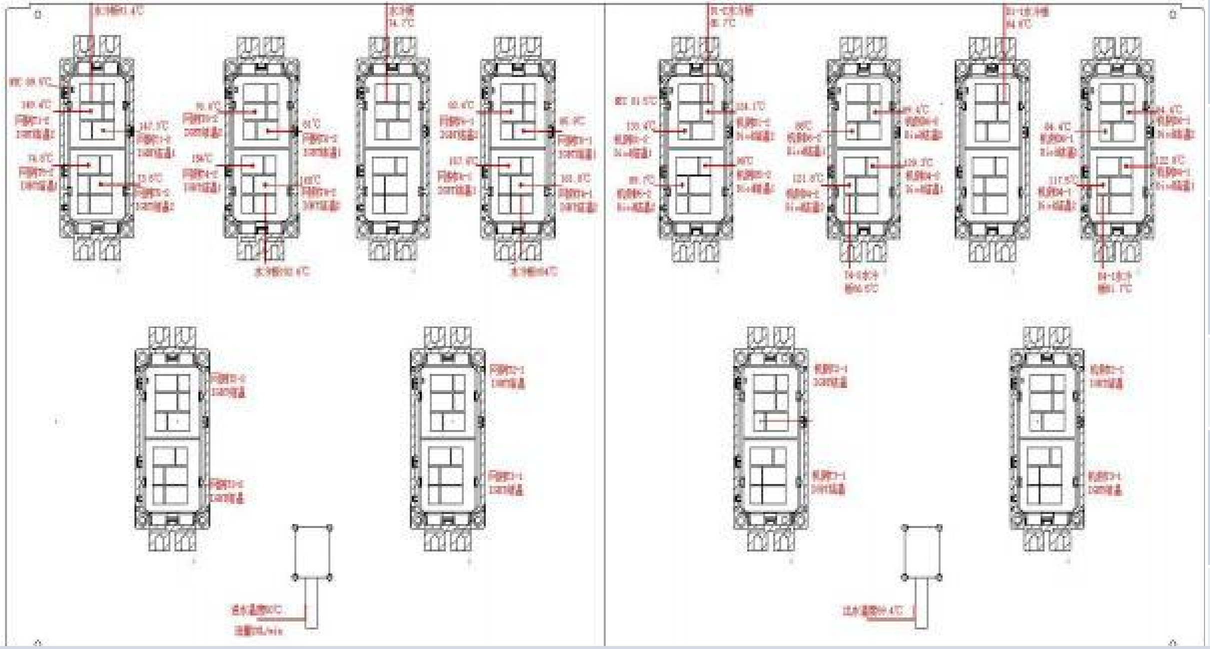
Liquid Cooling Solution for High-Power Electronics
Cold Plate for Ultra-High-Voltage Converter
Product Name: Cold Plate for Ultra-High-Voltage Converter
Heat Source: GTO
Power Dissipation: 2 x 2,500 W
Material: Aluminum
Welding Process: Friction Stir Welding (FSW)
Coolant: Pure water
Pressure Drop: 105 kPa at 6 L/min
Dimensions: 175 x 120 x 28 mm
Design Requirements:
1.Flow rate: 6 L/min
2.Coolant inlet temperature: 40°C
3.Cold plate temperature rise: ≤ 25°C



Liquid Cooling Solution for High-Power Electronics
Cold Plate for Rail Transit
Product Name: Rail Transit Cold Plate
Heat Source: 3 x IGBT Modules
Power Dissipation: 3,867 W
Material: Aluminum
Welding Process: Friction Stir Welding (FSW)
Coolant: 50% Glycol-Water Mixture
Pressure Drop: 18 kPa at 25 L/min
Dimensions: 390 x 380 x 25 mm
Design Requirements:
1.Flow Rate: 25 L/min
2.Coolant Inlet Temperature: 53°C
3.Cold Plate Temperature Rise: ≤ 30°C



Liquid Cooling Solution for High-Power Electronics
IGBT Cold Plate for 6.25 MW / 14 MW Converter Systems
Key Parameters:
•Heat Source: 12 IGBTs
•Total Power Dissipation: > 10,000 W
•TG600 Chip Temperature Limit: < 135°C
•Original Solution Temperature: Approximately 161.8°C
•ToneCooling Solution: Met the temperature requirement under the specified case conditions



NO.	Items	Technical Parameters
1	Heat Consumption	(529 + 365) x 12 = 10728W
2	Medium	53% Ethylene Glycol
3	Inlet Water Temperature	50 deg C
4	Flow Rate	20L/min
5	Flow Resistance	0.63 + / - 0.05 bar
6	Dimensions	636 x 378 x 16 mm

Industry	Customer Scope	Application scenarios	Heat-generating
Big computing	Data centers	Servers workstation	CPU GPU supply,wind-improving
	Servers	Servers, workstation	ASIC / AI GPU / DPU / Intel Modules， Storage / Hard Disk / Power Modules
	Switches	Switches	ASIC / Oil-Cooled Modules
	Graphics Card	Graphics Card	GPU
	Chips	Testing equipment	CPU, GPU,TEC
Automotive	Electric drive and control	Main controllers	IGBT
	Autonomous driving, intelligent cockpits	Domain controllers	CPU, GPU
	Rail transit	Traction converters	IGBT
Power grid Light	Wind power, photovoltaic power, energy storage	PCS	IGBT
	Power transmission and distribution	PCS， SVG	IGBT， IEGT， GTO
	Power Supply	Power Supply	IGBT

Products

Distribution Pipe

Liquid Cooling Expert for High Power Chips

Rack Manifolds

Liquid Cooling Expert on High Power Chips

Room Distribution Pipe

Designed according to system configuration and server room layout

Prefabricated, tested, and pre-assembled in the factory

On-site assembly support

Liquid Cooling Solutions for High-Power Electronics

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Mass production capacity

Production Process
Liquid Cooling Solutions for High-Power Electronics

Before Welding	Welding	After Welding	Downstream Processing
Material Purchase Milling Cutter CNC Cleaning	TLP Welding Induction Welding TIG Welding Vacuum Brazing FSW Welding Flame Welding	Cutting CNC	Surface Treatment Assembly Testing

Processing Equipment
Liquid Cooling Solutions for High-Power Electronics

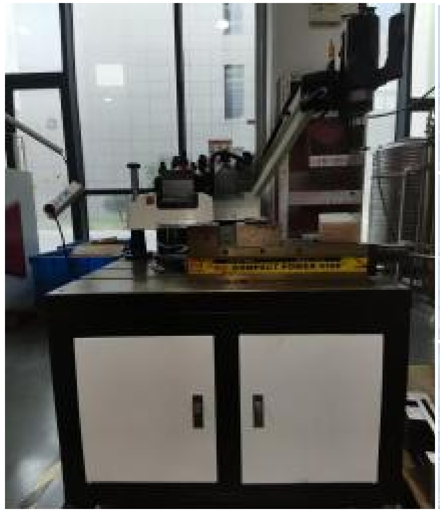
Name	Four-spindle Machining Center
Product Specification	MaximumL3000mm*W3000mm*H3000mm
Machining Precision	0.003mm
Rotational Speed	30000r/min
Stroke	X: 260mm Y: 350mm Z: 500mm

Name	Five - axis and Five - link Drilling and Milling Machine
Workbench Size	420mm*260mm
Operation Mode	Maximum control of five - axis, PLC control of five - axis
Feed Rate	0-150%, 16- level adjustment
Rotational Speed	100000r/min
Spindle Multiplier	50%-120%, eight - level adjustment

Name	Chamfering Machine
Product Specification	Maximum: L3300 x W1800 x H2150 mm
Model	Model: RC-25AF

Name	Sandblasting Machine
Model	TS-9060 Jinya Pressure Abrasive Blasting Machine
Purpose	Surface covering

Processing Equipment Welding equipment



Name	Electric Tapping Machine
Model	2022-D, M3 - 16, Rotation Speed 625
Purpose	Tapping threads for products

Name	Atmosphere BOX Resistance Furnace
Model	Welding Equipment
Purpose	Nitrogen Welding Furnace

Welding equipment Welding equipment

Name	Conveyor Brazing Furnace Atmosphere
Overall dimensions	L15800*W1750*H 1900mm
Maximum temperature	1050 deg C

Name	Friction Stir Welding Machine
Product Specificatio n	00 Maximum L1230 mm * W600mm * H 0mm
Forging Force	30KN
Welding Thickness	1-10mm

Inspection Equipment
Liquid Cooling Solutions for High-Power Electronics

Name	Ultrasonic scanner S800
Product Specification	Maximum L1950mm* W1520mm *H2000mm
Operating conditions	Pure water
Measurement error	+/-0.1%

Name	Flow Resistance Testing Machine
Workbench size	2600mm（L）x2230mm（W）x1620（H）mm
Test product properties	Copper/Aluminum
Flow rate	Small: 0.5 - 4 L/min Large: 4 - 15

Name	CTH Test Machine
Product Specification	Maximum: L1250 x W1250 x H1600 mm
Model	Model: LJ-225L
Temperature Range	-40 to 150 deg C
Humidity Range	20-98% RH

Name	Pressure Difference Testing Machine
Workbench Size	1200 mm (L) x 820 mm (W) x 1620 mm (H)
Product Test Properties	Applicable materials: Copper / Aluminum
product	

Inspection Equipment

Name	Helium Leak Detection Equipment
Maximum workpiece size	750mmX450mm X300mm
Nitrogen filling Pressure	0-1.0MPa (adjustable within this range)
Maximum internal cavity volume	<=3L

Liquid Cooling Expert for High Power Chips

Name	Integrated Detection Machine
Overall dimensions	L2000*W1800* H1800mm
Experimental medium	Pure water
Pressure holding time	Arbitrarily set

Cleaning Equipment

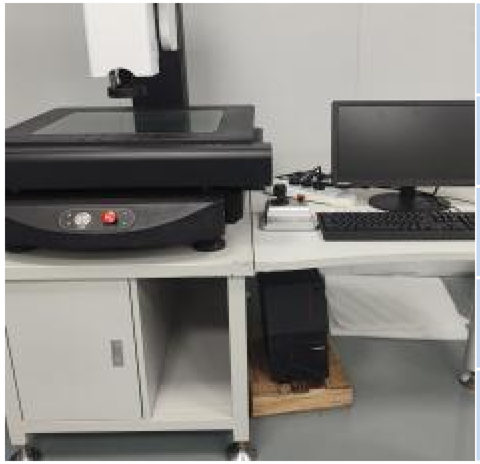
Name	Washing Machine
Overall dimensions	L8400*W2700* H3500mm
Cleaning cycle	5min/frame (the time is adjustable)
Total power of the equipment	Approximately 117KW, with an average power consumption of 60kW

Name	Cold Plate Cleaning and Leak Detection Machine
Category	Cleaning and Inntspection Equipme
Purpose	Cleaning the internal flow channels of cold plates

Other Equipments

Liquid Cooling Expert for High Power Chips

Name	Pure Water Equipment
Workbench Dimensions	3500mm (L) x1100mm (W) x2000 (H) mm
PH value	Between 5.7 and 7



Name	Video Measuring Instrument
Product Specification	Maximum L1500mm* W1040mm*H1840mm
Z-axis lifting and lowering stroke	150mm
Digital display resolution of XYZ	0.5
Video magnification	33X-195X

Name	Electric Thermostatic Forced Draft Drying oven
Category	Drying Equipment
Purpose	Drying

Name	Laser Marking Machine
Laser Wavelength	1064nm+/-5nm
Repeatability Accuracy	+0.003mm
Operation Purpose	Engrave QR codes on products to distinguish products and enable traceability
Height of Processed Products	Less than 400mm
Surface Requirement of Workpieces	The product surface should be flat
Processing Range	100mm*100mm (The range varies with different Lenses

Quality assurance capacity

Liquid Cooling Expert for High Power Chips

Product Inspection
Liquid Cooling Solutions for High-Power Electronics

Inspection Items	Technical Requirements	Quality Characteristics	Inspection Methods	Inspection Instruments
Appearance	Appearance	No burrs on the surface and edges, no bumps, scratches, oxidation or blackening defects	All-inspection	Visual inspection
Dimensional inspection	Flatness	Flatness within the range of 100*120mm <= 0.02mm	All-inspection	Coordinate measuring machine
	Roughness	Rz10, <=10um	All-inspection	Coordinate measuring machine
	Parallelism	0.04mm	All-inspection	Coordinate measuring machine
	Screw holes	The go gauge can pass through, and the no-go gauge cannot pass through	All-inspection	Go and no-go gauge
	Dimensions	For tolerances not specified in the drawing, followGB/T1804-f, GB/T1184-K	All-inspection	Vernier caliper

Product Inspection
Liquid Cooling Solutions for High-Power Electronics

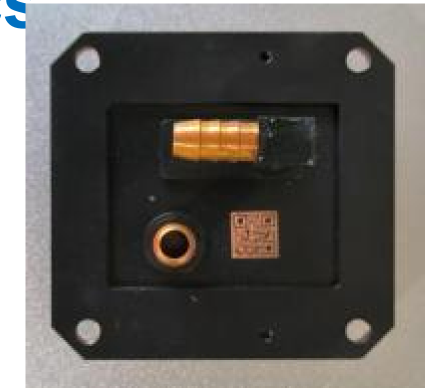
Inspection Items	Technical Requirements	Quality Characteristics	Inspection Methods	Instruments
Performance test	Ultrasonic testing	Welding joint efficiency>=85%	All-inspection	Ultrasonic scanning microscope
	Flow resistance testing	With a flow rate of 10L/min, the overall flow resistance is 40.5+/-2Kpa	All-inspection	Flow resistance detector
	Pressure testing	Apply a pressure of 1 MPa, hold the pressure for 5 minutes with no bulging, and the leakage amount is less than 3 Kpa.	All-inspection	Sealing tester
	Pressure-holding testing	Apply a pressure of 1 MPa, hold the pressure for 6 hours with no bulging, and the leakage amount is less than 30 Kpa.	All-inspection	Sealing tester
	Cleanliness testing	TDS is lower than 30 ppm.	All-inspection	Cleanliness detector
	Dryness testing	The humidity of the gas inside the product is <= 10%.	All-inspection	Dryness detector
Packaging inspection	Packing box inspection	Put one packing list inside each packing box, paste the packing list on the outside of the outer packing box. Each batch of products shall be accompanied by the inspection report corresponding to this batch of products.	All-inspection	Visual inspection

Quality System Documents
Liquid Cooling Solutions for High-Power Electronics

NO.	Document Number	Document Name
1	THS-QP-01	Document and Record Control Procedure
2	THS-QP-02	Management Review Control Procedure
3	THS-QP-03	Internal Audit Control Procedure
4	THS-QP-04	Human Resources Control Procedure
5	THS-QP-05	Contract Review Control Procedure
6	THS-QP-06	Customer Communication Control Procedure
7	THS-QP-07	Information Exchange Control Procedure
8	THS-QP-08	Design and Development Control Procedure
9	THS-QP-09	Procurement Process Control Procedure
10	THS-QP-10	Supplier Management Control Procedure
11	THS-QP-11	Engineering Change Control Procedure
12	THS-QP-12	Production Equipment Control Procedure
13	THS-QP-13	Production Process Control Procedure

No.	Document Number	Document Name
14	THS-QP-14	Protection and Delivery Control Procedure
15	THS-QP-15	Identification and Traceability Control Procedure
16	THS-QP-16	Product Monitoring and Measurement Control Procedure
17	THS-QP-17	Monitoring and Measuring Equipment Control Procedure
18	THS-QP-18	Nonconforming Product Control Procedure
19	THS-QP-19	Corrective and Preventive Control Procedure
20	THS-QP-20	Objective Management and Analysis Control Procedure
21	THS-QP-21	Continuous Improvement Control Procedure
22	THS-QP-22	Organizational Environment and Interested Parties Control Procedure
23	THS-QP-23	Risk and Opportunity Control Procedure
24	THS-QP-24	Knowledge Management Control Procedure
25	THS-QP-25	Customer and Supplier Property Control Procedure

Traceability System
Liquid Cooling Solutions for High-Power Electronics



A unique QR code is laser-engraved on each cold plate to track process and test results for individual cold plates and finished products, supporting product traceability and quality control.

Main monitoring and tracking points

Key Process Control

ection and Feedback

roduct Inspection and Shipment

Online detection points are set on the production line for real-time inspection. When issues are identified, feedback and adjustment actions are taken promptly.

Key processes such as welding and critical dimensions are strictly controlled. Performance tests, including C-scan inspection, airtightness testing, thermal performance testing, and flow-resistance testing, are conducted to confirm that the product meets the required standards. Original test data is retained for traceability.

Traceability System
Liquid Cooling Solutions for High-Power Electronics

Role of the Traceability System in Quality Management
The quality tracking system helps ensure that every stage of the cold plate process, from raw material procurement to finished product shipment, meets quality requirements. By recording and monitoring key production parameters, quality issues can be detected and corrected in a timely manner, supporting product reliability and safety.

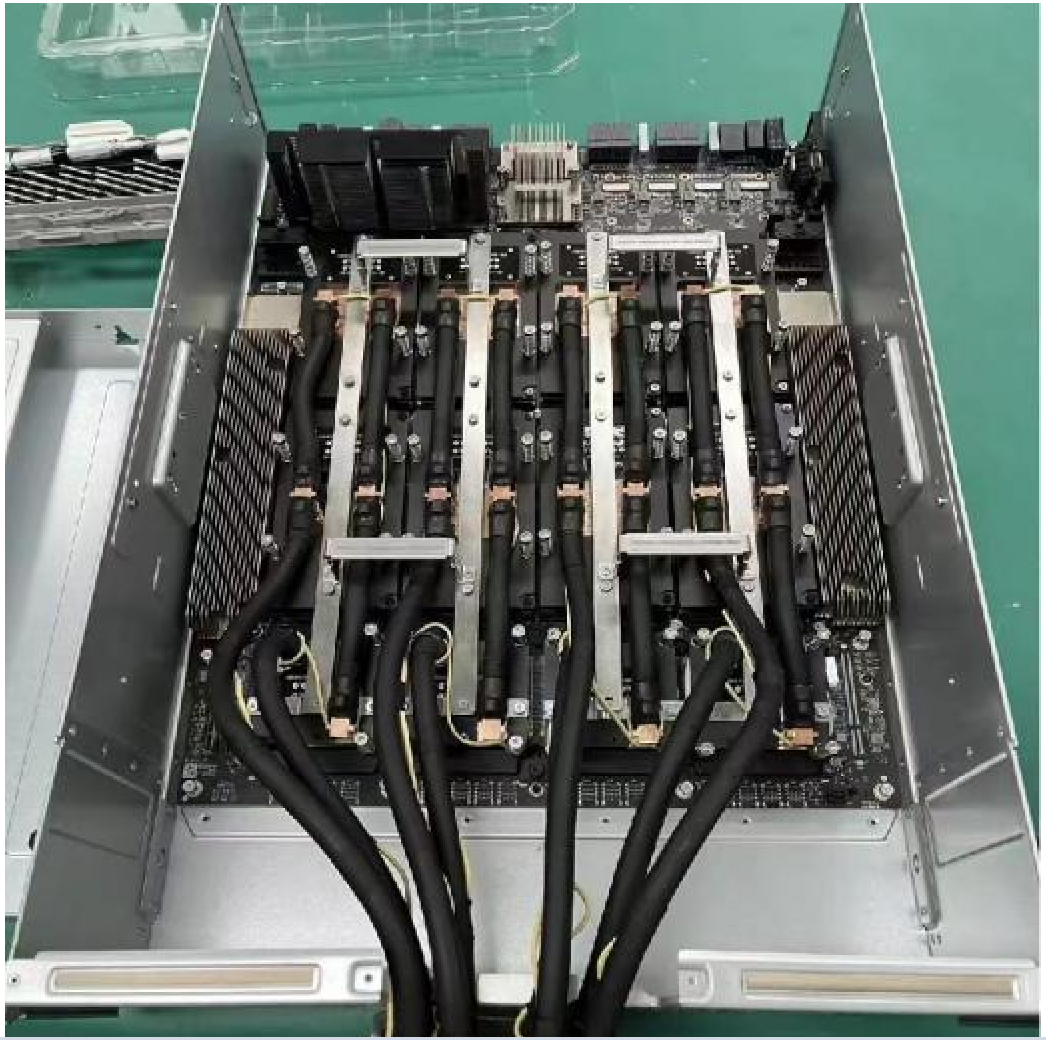
Impact of the Traceability System on Customer Trust
Through the quality tracking system, customers can access detailed production and inspection records for cold plates, which helps increase confidence in product quality.

Role of the Traceability System in Risk Control
The quality tracking system helps locate the source of problems quickly and supports effective corrective measures when product issues occur.

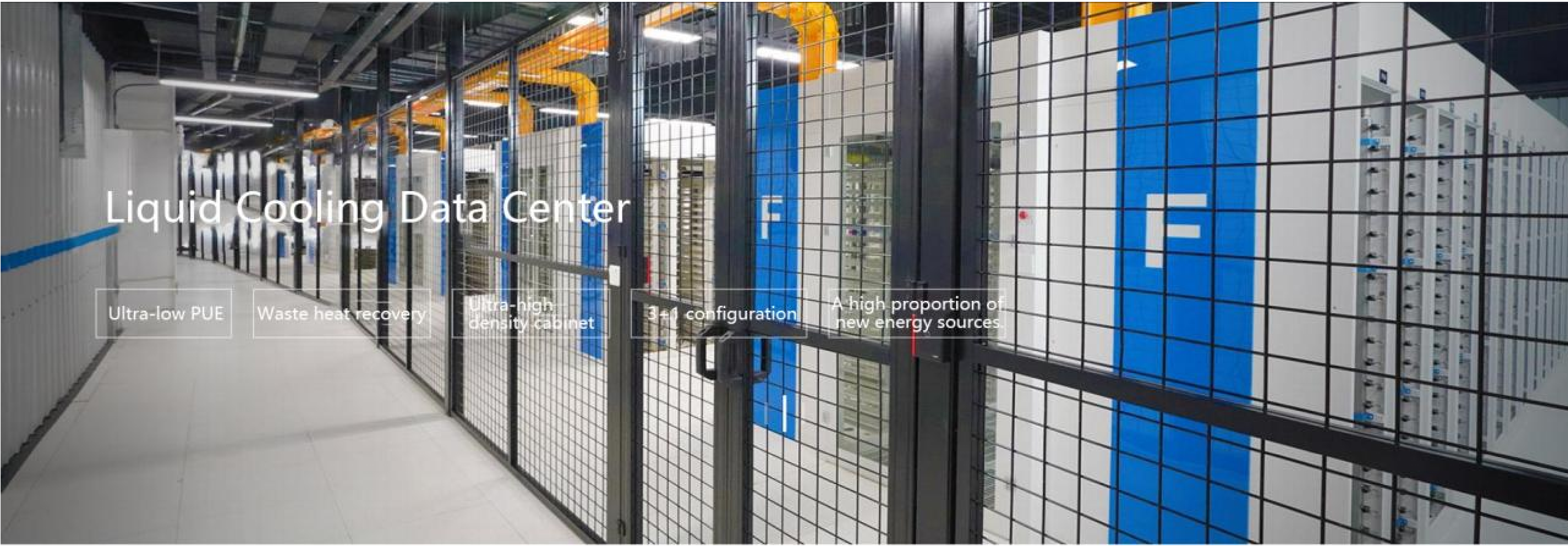
Prospects for cooperation

Liquid Cooling Expert for High Power Chips

Case: X Fusion AI Computing Server
Liquid Cooling Solutions for High-Power Electronics



Case: Data Center AI Computing Project
Liquid Cooling Solutions for High-Power Electronics



ToneCooling Advantages

Liquid Cooling Solutions for High-Power Electronics

Product Line and R&D

Rack-, cabinet-, and room-level CDU support

Cold plates for Intel, AMD, NVIDIA, and other chip platforms

Batch Supply and Quality

Practical validation and batch supply experience

Service before, during, and after sales

Engineering Support

Custom design, DFM review, sample validation, and production support

For customer RFQs, ToneCooling can support drawing review, process route selection, prototype validation, and production readiness discussion based on the final application requirements.

Thank You

Custom Liquid Cold Plates & Thermal Management Solutions

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Guangdong, China 516127

We look forward to supporting your next thermal management project.